

Learning Module: Archetypes in openEHR: A First Introductory Lesson

Section 1: Essential Vocabulary

Archetype:	A model, pattern, or template used to organize specific information.
Electronic Health Record (EHR):	A digital record that holds a person's medical history and information.
Clinical:	Related to the treatment of patients in a hospital or clinic.
Data:	Facts, numbers, or information used by computers.
Standard:	An agreed way of doing something so everyone understands it.
Constraint:	A rule or a limit that controls what you can write.
Template:	A large plan that groups small parts together for a specific use.
Observation:	The act of looking at something or measuring something to get information.
Evaluation:	Making a judgment or decision based on information.
Instruction:	An order or direction telling someone to do something.

Section 2: Information Article

In the modern world, hospitals use computers to save patient information. This is called an Electronic Health Record, or EHR. However, there is a big problem. Sometimes, one computer system cannot understand the data from another system. This is like two people speaking different languages. To fix this, we need a standard. One very important standard is called openEHR. The most important part of openEHR is the 'archetype'.

What is an Archetype?

Imagine you have a box of Lego blocks. Each block has a specific shape and size. You can use these blocks to build a house, a car, or a boat. In openEHR, an archetype is like a single Lego block. It is a model for one specific piece of clinical information. For example, there is an archetype for 'Blood Pressure'. There is another archetype for 'Body Temperature'.

An archetype tells the computer exactly what the information looks like. For 'Blood Pressure', the archetype says, 'We need two numbers: Systolic and Diastolic'. It also says, 'The unit is millimeters of mercury (mmHg)'. Because of this archetype, every computer knows exactly what 'Blood Pressure' means. They all agree on the pattern.

The Structure of Archetypes

Archetypes are very organized. They are not just random notes. They have a specific structure.

1. **Concept**: This is the name of the archetype, like 'Heart Rate'.
2. **Description**: This explains what the archetype is for.
3. **Definition**: This is the main part. It lists all the data points. For example, in a 'Weight' archetype, the data point is the number of kilograms.
4. **Rules (Constraints)**: These are limits. A rule might say, 'Weight cannot be a negative number'. This helps keep the data correct.

Classes of Archetypes

There are different types of archetypes for different jobs. These are called classes.

- **Observation**: This is for information we measure or see. Examples are Blood Pressure, Weight, or color of the skin.
- **Evaluation**: This is for when a doctor thinks about the data and makes a decision. A

'Diagnosis' (saying what sickness a person has) is an evaluation.

- **Instruction**: This is when a doctor tells a nurse or patient to do something. For example, 'Take this medicine'.
- **Action**: This records that something was done. For example, 'The patient took the medicine'.

Templates

An archetype is just one piece of the puzzle. When a doctor sees a patient, they need many pieces. They need to check weight, temperature, and ask questions. To do this, we put many archetypes together. This group is called a 'Template'. A template is like a full form that the doctor fills out. It uses the archetypes as building blocks.

Why use Archetypes?

Using archetypes is very helpful. First, it helps computers share data. Second, archetypes can be translated. The computer code stays the same, but the words can change from English to Spanish or French. This means doctors all over the world can use the same models. openEHR archetypes make medical data safe, clear, and easy to understand.

Section 3: Reading Comprehension Quiz

1. What does EHR stand for?

Difficulty: EASY

- A) Electric Home Robot
- B) Electronic Health Record
- C) Easy Health Rules
- D) English Hospital Room

2. What is an archetype compared to in the text?

Difficulty: EASY

- A) A puzzle piece.
- B) A Lego block.
- C) A brick wall.
- D) A fast car.

3. Which of the following is an Observation?

Difficulty: EASY

- A) Taking medicine.
- B) A diagnosis.
- C) Body Temperature.
- D) A prescription.

4. What are the rules in an archetype called?

Difficulty: EASY

- A) Constraints
- B) Laws
- C) Orders
- D) Blocks

5. What do we call a group of archetypes?

Difficulty: EASY

- A) A Standard
- B) A Template
- C) A Class
- D) A Constraint

6. Why is it a problem if computers do not have a standard?

Difficulty: INTERMEDIATE

- A) The computers will turn off.
- B) The computers cannot understand each other's data.
- C) The doctors will not get paid.
- D) The internet will break.

7. Based on the text, what is the difference between Observation and Evaluation?

Difficulty: INTERMEDIATE

- A) Observation is seeing facts; Evaluation is making a decision.
- B) Observation is for nurses; Evaluation is for patients.
- C) Observation is digital; Evaluation is paper.
- D) There is no difference.

8. What is the function of the 'Definition' part of an archetype?

Difficulty: INTERMEDIATE

- A) It names the archetype.
- B) It lists all the data points needed.
- C) It translates the language.
- D) It saves the file.

9. Why are templates useful for a doctor's visit?

Difficulty: INTERMEDIATE

- A) They make the computer faster.
- B) They combine all the necessary archetypes into one form.
- C) They delete old information.
- D) They prevent the doctor from typing.

10. What benefit does openEHR offer for international hospitals?

Difficulty: INTERMEDIATE

- A) Free computers.
- B) Archetypes can be translated so data is understood globally.
- C) Doctors can travel for free.
- D) It forces everyone to speak English.

Section 4: Reading Quiz Answer Key

Question 1: Electronic Health Record Question 2: A Lego block.

Question 3: Body Temperature. Question 4: Constraints

Question 5: A Template Question 6: The computers cannot understand each other's data.

Question 7: Observation is seeing facts; Evaluation is making a decision. Question 8: It lists all the data points

Question 9: They combine all the necessary archetypes into one form. Question 10: Archetypes can be translated

Section 5: Dialogue Transcript (Listening)

- Joe:** Hi Jane. Do you have a moment? I am really confused about something.
- Jane:** Hi Joe. Sure, I have time. What is confusing you today?
- Joe:** Well, I started my new job in the hospital IT department. My boss told me I need to learn about 'openEHR' and something called 'archetypes'. I have no idea what that means.
- Jane:** Ah, I see! Don't worry, Joe. It sounds difficult, but it is actually quite logical. I can help you explain it.
- Joe:** That would be great. Can we start with the basics? What is openEHR?
- Jane:** Okay. openEHR is a way to organize medical data. You know that doctors write notes about patients, right?
- Joe:** Yes, they write things like blood pressure, allergies, and medicines.
- Jane:** Exactly. In the past, every hospital used a different way to write these notes. One hospital might write 'BP' for blood pressure. Another might write 'Blood Press'. A computer cannot understand that 'BP' and 'Blood Press' are the same thing.
- Joe:** Oh, I understand. So the computers get confused.
- Jane:** Yes. So, openEHR provides a standard. It gives us a set of rules so all computers understand the data. The most important tool for this is the 'archetype'.
- Joe:** Okay, so what is an archetype exactly? Is it a computer program?
- Jane:** Not exactly. Think of it as a model or a pattern. Do you like Lego blocks?
- Joe:** Yes, I love building with Legos.
- Jane:** Great. Imagine that clinical data is like a building. An archetype is like a specific type of Lego block. One archetype defines one specific concept. For example, there is an archetype for 'Blood Pressure'.
- Joe:** Okay. So the Blood Pressure archetype is a block. What is inside this block?
- Jane:** Inside the archetype, there is a definition. It tells us what data we need. For blood pressure, we need two numbers. Do you know what they are?
- Joe:** I think so. Systolic and diastolic?
- Jane:** Correct! The archetype says: 'You must have a place for Systolic and a place for Diastolic'. It also says what units to use, like millimeters of mercury.
- Joe:** Why is that important? Can't the doctor just type the numbers?
- Jane:** If the doctor just types numbers, the computer doesn't know what they mean. The archetype gives the numbers a label and a meaning. It makes the data structured.

Joe: I see. Does the archetype do anything else besides labeling?

Jane: Yes, it has rules. In openEHR, we call these 'constraints'. A constraint is a limit.

Joe: What kind of limit?

Jane: Well, think about body temperature. Can a human have a body temperature of 1000 degrees?

Joe: No, that is impossible! We would burn up.

Jane: Exactly. So, the archetype has a constraint. It might say, 'Temperature must be between 20 degrees and 45 degrees'. If a doctor types 1000, the computer will say 'Error! This is wrong'.

Joe: That is very smart. It stops mistakes.

Jane: Yes. Now, you need to know that there are different classes of archetypes. Not all medical data is the same.

Joe: What do you mean by classes?

Jane: We group them by what they do. The first class is called 'Observation'. This is the most common one.

Joe: Observation... that means looking at something, right?

Jane: Yes. It is information that you can measure or see. Blood pressure, weight, height, and heart rate are all Observations. You observe the patient to get the number.

Joe: Okay, that makes sense. What is the next class?

Jane: The next one is 'Evaluation'. This is different. An evaluation is not just a number. It is an opinion or a decision made by the doctor.

Joe: Can you give me an example?

Jane: Sure. A diagnosis is an evaluation. If the doctor looks at your symptoms and says, 'You have the flu', that is an Evaluation. It comes from the doctor's brain, not just a machine.

Joe: I understand. Observation is a fact, Evaluation is a decision.

Jane: Perfect, Joe! You are learning fast. The third class is 'Instruction'. This is when the doctor tells someone to do something.

Joe: Like giving a prescription?

Jane: Yes! 'Take two pills every morning' is an Instruction. It is an order.

Joe: And is there a class for when I actually take the pills?

Jane: Yes, that is the fourth class. It is called 'Action'. An Action records what actually happened. 'Joe took the pill at 8 AM'.

Joe: So we have Observation, Evaluation, Instruction, and Action. It covers everything.

Jane: It covers almost everything in clinical care. But Joe, one archetype is usually not enough for a doctor's visit.

Joe: Why not?

Jane: Because when you go to the doctor, they check many things. They check your heart, your weight, and they ask how you feel. That is three or four archetypes.

Joe: So how do we put them together?

Jane: We use something called a 'Template'. A Template is a group of archetypes.

Joe: Is it like a form?

Jane: Yes, exactly. Imagine a form for a 'Yearly Checkup'. This template will include the Blood Pressure archetype, the Weight archetype, and the Height archetype. It groups them together for a specific purpose.

Joe: Can we change the archetypes when we put them in a template?

Jane: We cannot change the definition, but we can make it more specific. For example, an archetype might have a list of 100 possible symptoms. But for a 'Flu Check' template, we might only show the 5 symptoms related to the flu.

Joe: That is very useful. It makes the screen cleaner for the doctor.

Jane: Yes. And there is one more amazing thing about archetypes. They work in many languages.

Joe: Really? How?

Jane: The computer code—the structure—is the same everywhere. But the labels can be translated. So, a doctor in Spain sees 'Presión arterial', and a doctor in the UK sees 'Blood Pressure'. But the computer knows it is the exact same archetype.

Joe: Wow. That means we can share research and data across countries.

Jane: Exactly. That is the goal of openEHR. To make health data work together globally.

Joe: Thank you so much, Jane. This makes a lot of sense now. It is like building with Lego blocks to make sure everyone understands the structure.

Jane: You've got it, Joe. Good luck with your new job!

Joe: Thanks! I feel much more confident now.

Section 6: Listening Comprehension Quiz

1. Why is Joe asking Jane for help?

- A) He is sick and needs a doctor.
- B) He is new to the IT department and needs to learn about openEHR.
- C) He wants to buy Lego blocks.
- D) He lost his computer.

2. What analogy (comparison) does Jane use to explain archetypes?

- A) Cars on a road.
- B) Ingredients in a cake.
- C) Lego blocks.
- D) Pages in a book.

3. Which medical example does Jane use first to explain an archetype?

- A) Broken leg.
- B) Headache.
- C) Blood Pressure.
- D) Eye color.

4. According to the dialogue, what two numbers are needed for blood pressure?

- A) Height and Weight.
- B) Systolic and Diastolic.
- C) Fast and Slow.
- D) Hot and Cold.

5. Why does Jane say we cannot just type random numbers?

- A) Because the keyboard is broken.
- B) Because the computer needs labels to know what the numbers mean.
- C) Because doctors don't like numbers.
- D) Because numbers are boring.

6. What is a 'constraint' in openEHR?

- A) A type of medicine.
- B) A rule or limit on the data.
- C) A computer virus.
- D) A new hospital.

7. What example of a constraint does Jane give?

- A) You cannot eat before surgery.
- B) You must pay the bill.
- C) Body temperature cannot be 1000 degrees.
- D) Blood pressure must be high.

8. What is the 'Observation' class used for?

- A) Things you can measure or see.
- B) Giving orders.
- C) Making a guess.
- D) Doing an action.

9. Which of these is an example of an 'Evaluation'?

- A) Measuring height.
- B) Checking weight.
- C) A doctor making a diagnosis.
- D) Taking a pill.

10. What is the 'Instruction' class used for?

- A) Writing a history.
- B) Telling someone to do something (an order).
- C) Looking at a patient.
- D) Saving a file.

11. What does the 'Action' class record?

- A) What the doctor hopes will happen.
- B) What actually happened (like taking a pill).
- C) The cost of the visit.
- D) The patient's name.

12. What is a 'Template'?

- A) A single archetype.
- B) A group of archetypes put together for a specific use.
- C) A computer screen.
- D) A type of medicine.

13. Can you change the definition of an archetype inside a template?

- A) Yes, you can change everything.
- B) No, but you can make it more specific (hide things you don't need).
- C) Yes, you can delete the rules.
- D) No, templates cannot be changed.

14. What does Jane say about languages?

- A) Archetypes only work in English.
- B) Archetypes only work in Spanish.
- C) The structure is the same, but labels can be translated.
- D) You need a new computer for every language.

15. How does Joe feel at the end of the conversation?

- A) More confused.
- B) Angry.
- C) Hungry.
- D) More confident.

Section 7: Listening Quiz Answer Key

Q1: He is new to the IT department and needs to learn about openEHR. | Q2: Lego blocks. | Q3: Blood Pressure. | Q4: Systolic and Diastolic. | Q5: Because the computer needs labels to know what the numbers mean. | Q6: A rule or limit on the data. | Q7: Body temperature cannot be 1000 degrees. | Q8: Things you can measure or see. | Q9: A doctor making a diagnosis. | Q10: Telling someone to do something (an order). | Q11: What actually happened (like taking a pill). | Q12: A group of archetypes put together for a specific use. | Q13: No, but you can make it more specific (hide things you don't need). | Q14: The structure is the same, but labels can be translated. | Q15: More confident.

Teacher's Overall Feedback:

Dialogue Q1: He is new to the IT department and needs to learn about openEHR. Dialogue Q2: Lego blocks. Dialogue Q3: Blood Pressure. Dialogue Q4: Systolic and Diastolic. Dialogue Q5: Because the computer needs labels to know what the numbers mean. Dialogue Q6: A rule or limit on the data. Dialogue Q7: Body temperature cannot be 1000 degrees. Dialogue Q8: Things you can measure or see. Dialogue Q9: A doctor making a diagnosis. Dialogue Q10: Telling someone to do something (an order). Dialogue Q11: What actually happened (like taking a pill). Dialogue Q12: A group of archetypes put together for a specific use. Dialogue Q13: No, but you can make it more specific (hide things you don't need). Dialogue Q14: The structure is the same, but labels can be translated. Dialogue Q15: More confident. Textbook Q1: Electronic Health Record Textbook Q2: A Lego block. Textbook Q3: Body Temperature. Textbook Q4: Constraints Textbook Q5: A Template Textbook Q6: The computers cannot understand each other's data. Textbook Q7: Observation is seeing facts; Evaluation is making a decision. Textbook Q8: It lists all the data points needed. Textbook Q9: They combine all the necessary archetypes into one form. Textbook Q10: Archetypes can be translated so data is understood globally.